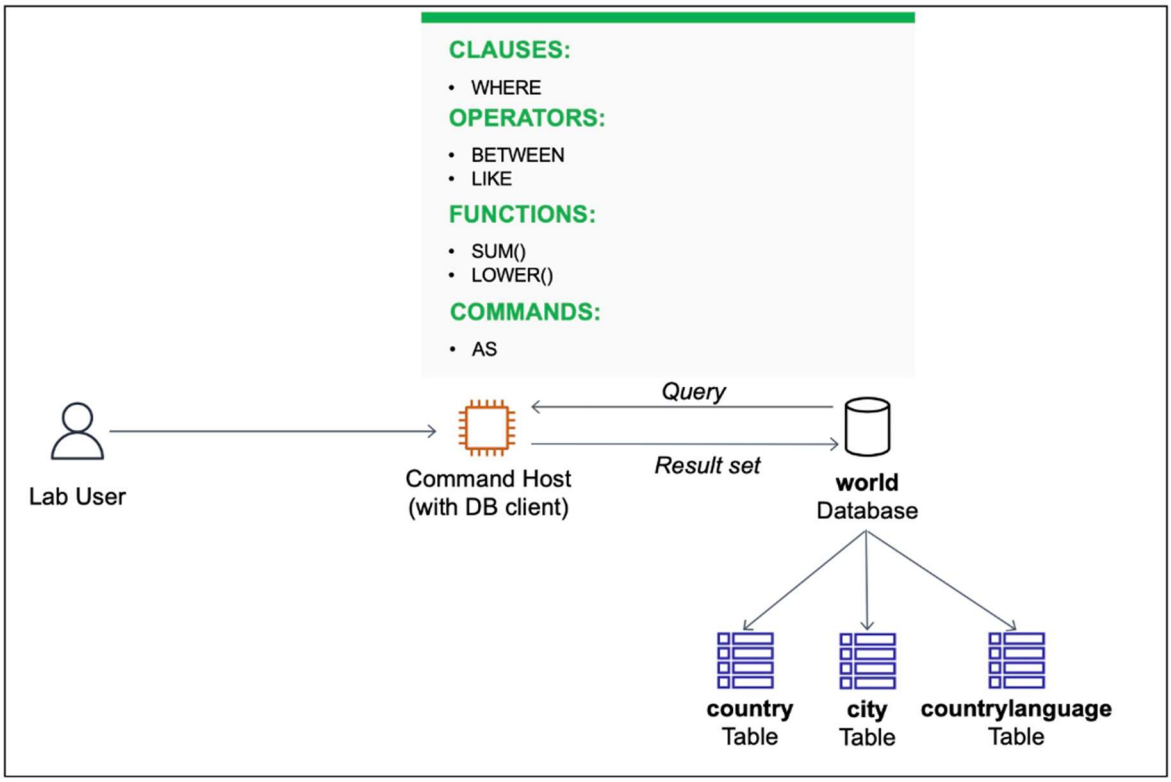
271--Lab - Realización de una búsqueda condicional

En esta práctica se muestra el funcionamiento de los siguientes comandos en SQL:

- WHERE para filtrar el resultado de una consulta
- BETWEEN para indicar el rango de una consulta
- LIKE para filtrar el resultado de una consulta con base a un patrón
- AS para definir alias para las columnas

Escenario

Se cuenta con una instancia EC2 configurada con un cliente de base de datos MySQL.



Para conectarnos a la instancia y a la base de datos seguimos los pasos de las prácticas anteriores.

Hacer consultas a la base de datos

Verificamos si la base de datos world está disponible

```
MariaDB [(none)]> SHOW DATABASES;
+--------------------+
| Database           |
+--------------------+
| information_schema |
| mysql              |
| performance_schema |
| world              |
+--------------------+
4 rows in set (0.001 sec)
```

Con el uso de *Where* y operadores de comparación consultamos los países con una población mayor o igual a 50,000,000 y menor o igual a 100,000,000

```
MariaDB [(none)]> SELECT Name, Capital, Region, SurfaceArea, Population FROM world.country WHERE Population >= 50000000 AND Population <= 100000000;
+-------------------------------------+---------+--------------------------+-------------+------------+
| Name                                | Capital | Region                   | SurfaceArea | Population |
+-------------------------------------+---------+--------------------------+-------------+------------+
| Congo, The Democratic Republic of the |    2298 | Central Africa           |  2344858.00 |   51654000 |
| Germany                             |    3068 | Western Europe           |   357022.00 |   82164700 |
| Egypt                               |     608 | Northern Africa          |  1001449.00 |   68470000 |
| Ethiopia                            |     756 | Eastern Africa           |  1104300.00 |   62565000 |
| France                              |    2974 | Western Europe           |   551500.00 |   59225700 |
| United Kingdom                      |     456 | British Islands          |   242900.00 |   59623400 |
| Iran                                |    1380 | Southern and Central Asia |  1648195.00 |   67702000 |
| Italy                               |    1464 | Southern Europe          |   301316.00 |   57680000 |
| Mexico                              |    2515 | Central America          |  1958201.00 |   98881000 |
| Philippines                         |     766 | Southeast Asia           |   300000.00 |   75967000 |
| Thailand                            |    3320 | Southeast Asia           |   513115.00 |   61399000 |
| Turkey                              |    3358 | Middle East              |   774815.00 |   66591000 |
| Ukraine                             |    3426 | Eastern Europe           |   603700.00 |   50456000 |
| Vietnam                             |    3770 | Southeast Asia           |   331689.00 |   79832000 |
+-------------------------------------+---------+--------------------------+-------------+------------+
14 rows in set (0.000 sec)
```

El resultado anterior también puede obtenerse mediante el uso de *BETWEEN AND*

```
MariaDB [(none)]> SELECT Name, Capital, Region, SurfaceArea, Population FROM world.country WHERE Population BETWEEN 50000000 AND 100000000;
+-------------------------------------+---------+--------------------------+-------------+------------+
| Name                                | Capital | Region                   | SurfaceArea | Population |
+-------------------------------------+---------+--------------------------+-------------+------------+
| Congo, The Democratic Republic of the |    2298 | Central Africa           |  2344858.00 |   51654000 |
| Germany                             |    3068 | Western Europe           |   357022.00 |   82164700 |
| Egypt                               |     608 | Northern Africa          |  1001449.00 |   68470000 |
| Ethiopia                            |     756 | Eastern Africa           |  1104300.00 |   62565000 |
| France                              |    2974 | Western Europe           |   551500.00 |   59225700 |
| United Kingdom                      |     456 | British Islands          |   242900.00 |   59623400 |
| Iran                                |    1380 | Southern and Central Asia |  1648195.00 |   67702000 |
| Italy                               |    1464 | Southern Europe          |   301316.00 |   57680000 |
| Mexico                              |    2515 | Central America          |  1958201.00 |   98881000 |
| Philippines                         |     766 | Southeast Asia           |   300000.00 |   75967000 |
| Thailand                            |    3320 | Southeast Asia           |   513115.00 |   61399000 |
| Turkey                              |    3358 | Middle East              |   774815.00 |   66591000 |
| Ukraine                             |    3426 | Eastern Europe           |   603700.00 |   50456000 |
| Vietnam                             |    3770 | Southeast Asia           |   331689.00 |   79832000 |
+-------------------------------------+---------+--------------------------+-------------+------------+
14 rows in set (0.000 sec)
```

Con *LIKE* y el comodín % consultamos los países que en la región contengan "Europe"

```
MariaDB [(none)]> SELECT sum(Population) from world.country WHERE Region LIKE "%Europe%";
+------------------+
| sum(Population)  |
+------------------+
|        634947800 |
+------------------+
1 row in set (0.000 sec)
```

Consultamos la suma de la población y definimos un alias con *AS* para el encabezado de la columna resultante

```
MariaDB [(none)]> SELECT sum(population) as "Europe Population Total" from world.country WHERE region LIKE "%Europe%";
+-------------------------+
| Europe Population Total |
+-------------------------+
|               634947800 |
+-------------------------+
1 row in set (0.001 sec)
```

Por lo regular las bases de datos están configuradas para ser sensibles a mayúsculas y minúsculas. Por tal razón, cuando se comparan cadenas de texto, una condición podría no cumplirse.

Para evitar las diferencias se pueden usar las funciones *LOWER()* y *UPPER() en la clausula WHERE*

```
MariaDB [(none)]> SELECT Name, Capital, Region, SurfaceArea, Population from world.country WHERE LOWER(Region) LIKE "%central%";
+-------------------------------------+---------+-------------------------+-------------+------------+
| Name                                | Capital | Region                  | SurfaceArea | Population |
+-------------------------------------+---------+-------------------------+-------------+------------+
| Afghanistan                         |       1 | Southern and Central Asia |   652090.00 |   22720000 |
| Angola                              |      56 | Central Africa          |  1246700.00 |   12878000 |
| Bangladesh                          |     150 | Southern and Central Asia |   143998.00 |  129155000 |
| Belize                              |     185 | Central America         |    22696.00 |     241000 |
| Bhutan                              |     192 | Southern and Central Asia |    47000.00 |    2124000 |
| Central African Republic            |    1889 | Central Africa          |   622984.00 |    3615000 |
| Cameroon                            |    1804 | Central Africa          |   475442.00 |   15085000 |
| Congo, The Democratic Republic of the |    2298 | Central Africa          |  2344858.00 |   51654000 |
| Congo                               |    2296 | Central Africa          |   342000.00 |    2943000 |
| Costa Rica                          |     584 | Central America         |    51100.00 |    4023000 |
| Gabon                               |     902 | Central Africa          |   267668.00 |    1226000 |
```

Por último, obtenemos la suma del área de superficie y población de los países donde la región contiene "North America"

```
MariaDB [(none)]> SELECT SUM(SurfaceArea) AS "Surface Area", SUM(Population) AS Population FROM world.country WHERE Region LIKE '%North America%';
+--------------+------------+
| Surface Area | Population |
+--------------+------------+
|  21500515.00 |  309632000 |
+--------------+------------+
1 row in set (0.000 sec)
```